Responsible AI (RAI) Framework Project Introduction

NovaCare Health Systems

Course: Responsible Appl of AI (AIML-510-01R4A)
Assignment: 1.4 Responsible AI (RAI) Framework Project Introduction
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Student Name	Student ID	Contribution
Vaibhavi Chirra	2926738	Researched and developed the fictional company's background, including its industry, size, operations, and need for a Responsible AI program.
Syeda Anees Fatima	2920992	Identified and explained the company's proposed AI use cases, including their business benefits and potential ethical risks.
Arshi Khan	2926773	Researched the AI ethical principles adopted by five major companies and gathered reliable sources for comparison.
Bhuvaneshwari Bibi Reddy	2935275	Created the comparison chart showing the common and unique AI ethical principles followed by the five companies.
Sugyeong Hong	2921747	Developed the recommended ethical principles for the fictional company and explained why common principles were included or excluded.
Tisungane Namacha	2921572	Drafted the company's AI ethics statement using guidance and best practices from Chapter 5 of *Ethical Machines*.
Laxmi Lahari Naidu Bugata	2930252	Reviewed the DoD Responsible AI Strategy and identified suitable methods and metrics for measuring AI ethics performance.
Akhila Chittimella	2925412	Prepared the conclusion, summarized the difficulties and questions raised by the workshop, and reviewed the entire presentation for formatting, consistency, citations, and accuracy.

Framework Context and Objectives

Continuing the NovaCare healthcare case study from Workshop 1.3

NovaCare is a large U.S. healthcare network using AI for medical imaging, patient-risk prediction, clinical documentation, and operational optimization.

RAI principles established in Workshop 1.3: patient safety & reliability, fairness & equity, privacy & security, transparency & explainability, human oversight, accountability, and governability.

Framework objective: convert those principles into accountable roles, repeatable controls, measurable KPIs, and lifecycle governance.

NIST AI RMF structure used as the operating backbone: GOVERN, MAP, MEASURE, and MANAGE.

Risk-based approach: higher-impact clinical AI receives stronger review, validation, monitoring, and executive accountability.

Top Seven RAI Roles – Leadership and Control

1. Chief Responsible AI Officer
Enterprise accountability; AI governance, policy, executive communication, risk management, healthcare regulation.

2. RAI Governance & Ethics Lead
Ethics reviews, policy interpretation, stakeholder analysis, facilitation, documentation, conflict resolution.

3. AI Risk & Compliance Manager
NIST AI RMF, HIPAA/privacy awareness, risk assessments, audit readiness, vendor risk, regulatory monitoring.

4. AI Safety & Validation Lead
Model validation, clinical safety, testing design, robustness, drift analysis, failure-mode assessment.

Top Seven RAI Roles – Technical and Stakeholder Functions

5. Data Governance & Privacy Lead
Data lineage, consent, minimization, access control, quality, retention, de-identification, privacy impact assessment.

6. AI Security & Red Team Lead
Threat modeling, adversarial testing, secure ML, access security, incident response, third-party AI security evaluation.

7. Human Factors & Clinical AI Lead
Clinical workflow, usability, explainability needs, human oversight, training, patient/clinician feedback, change management.

Cross-functional support: legal counsel, procurement, internal audit, cybersecurity operations, data science, clinical leadership, and patient representatives.

Responsible AI Organization and Accountability

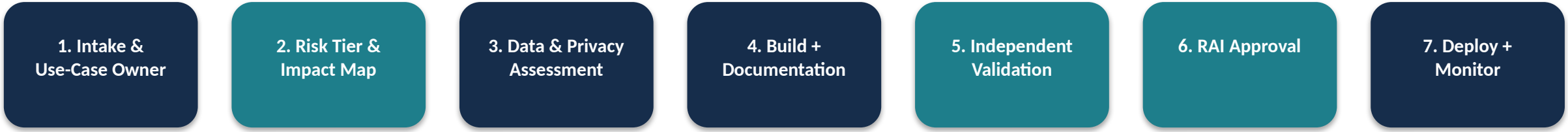


Accountable: Executive RAI Steering Committee and business/clinical owner for each AI system.

Responsible: product/model owner plus assigned RAI control leads.

Independent challenge: validation, compliance, security, privacy, and internal audit.

RAI Vetting Process – Internally Developed AI



Required evidence: intended purpose, affected populations, data lineage, model card, limitations, acceptance criteria, human-oversight design, security controls, and rollback plan.

Validation gates: performance and robustness, subgroup fairness, privacy/security, explainability/usability, clinical safety, and regulatory/compliance review.

High-risk clinical AI cannot self-approve; it requires independent validation plus RAI Steering Committee authorization.

Post-deployment monitoring includes drift, incidents, subgroup performance, override behavior, complaints, and periodic re-approval.

RAI Vetting Process – Externally Procured AI

1. Business need & risk classification — define intended use, decision impact, users, patient exposure, and alternatives.

2. Vendor due diligence — review training-data practices, model documentation, security, privacy, certifications, incident history, and subcontractors.

3. Contractual safeguards — require audit rights, breach/incident notification, change notification, data-use limits, service levels, documentation, and termination/exit provisions.

4. Independent NovaCare testing — validate performance on representative local data; test fairness, robustness, usability, security, and integration risks.

5. Approval & controlled pilot — limit initial deployment, define human oversight, collect user feedback, and confirm KPI thresholds.

6. Ongoing vendor/model monitoring — reassess after material model updates, data changes, incidents, performance drift, or regulatory changes.

Procurement does not transfer accountability to the vendor.

12-Month RAI Organization Implementation Timeline

Months 1–2	Foundation	Executive sponsor; RAI charter; inventory AI systems; define principles and risk tiers.
Months 3–4	Organization	Appoint seven RAI roles; form steering committee; define RACI and escalation paths.
Months 5–6	Controls	Create internal/vendor vetting standards, impact assessments, documentation templates, and approval gates.
Months 7–8	Pilot	Apply framework to 2–3 AI use cases; train teams; test dashboards and incident process.
Months 9–10	Scale	Extend governance to all AI projects/vendors; integrate procurement, security, privacy, and audit.
Months 11–12	Optimize	Audit effectiveness; review KPI trends; close gaps; publish annual RAI report and Year-2 roadmap.

Actionable Responsible AI KPIs

Principle	KPI	Target / Trigger	Management Action
Safety & Reliability	Critical AI incidents; validation pass rate	0 severe incidents; ≥95% pass	Suspend/rollback; root-cause review
Fairness & Equity	Max subgroup performance gap	≤5 percentage points	Bias investigation; retrain/restrict
Privacy & Security	AI privacy/security findings	0 critical; 100% closed on time	Block release or remediate
Transparency	High-risk models with current model cards	100%	Escalate missing/expired documentation
Human Oversight	Clinician override + inappropriate automation rate	Trend within approved band	Workflow review and retraining
Accountability	AI systems with named owner & current review	100%	No owner = no production use
Governability	Time to disable unsafe AI after confirmed trigger	≤30 minutes for critical systems	Test kill switch and escalation quarterly

KPI design: named owner + measurable threshold + review cadence + predefined corrective action.

KPI Review and Escalation Model

Weekly: operational model monitoring — drift, outages, alerts, overrides, and security events.

Monthly: product owner + RAI control review — KPI exceptions, subgroup performance, complaints, and corrective actions.

Quarterly: RAI Steering Committee — portfolio risk, high-risk model re-approval, vendor changes, audit findings, and trend analysis.

Annually: Board Risk / Quality Committee — RAI effectiveness, material incidents, ethical-principle alignment, resources, and framework updates.

Escalation rule: threshold breach creates a named action owner and deadline; critical safety/privacy/security triggers can automatically pause deployment.

Measure → Compare to Threshold → Assign Action → Verify Closure

Conclusion

NovaCare's RAI framework turns ethical principles into an operating system for AI governance.

Seven defined roles create clear ownership while independent validation prevents product teams from approving their own high-risk systems.

Separate internal-development and vendor-procurement processes address different sources of AI risk while maintaining the same ethical standards.

The 12-month roadmap starts with governance foundations, pilots controls, then scales them across the enterprise.

Actionable KPIs connect each ethical principle to a measurable threshold and a management response—making Responsible AI auditable and sustainable.

References

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Ethical Machines, Chapter 6.

NovaCare Health Systems is a fictional organization created for this course case study.